Agentic Architecture Exploration

The proposed framework connects AI workloads, compiler decisions, architecture evaluation, and physical implementation in one iterative workflow:

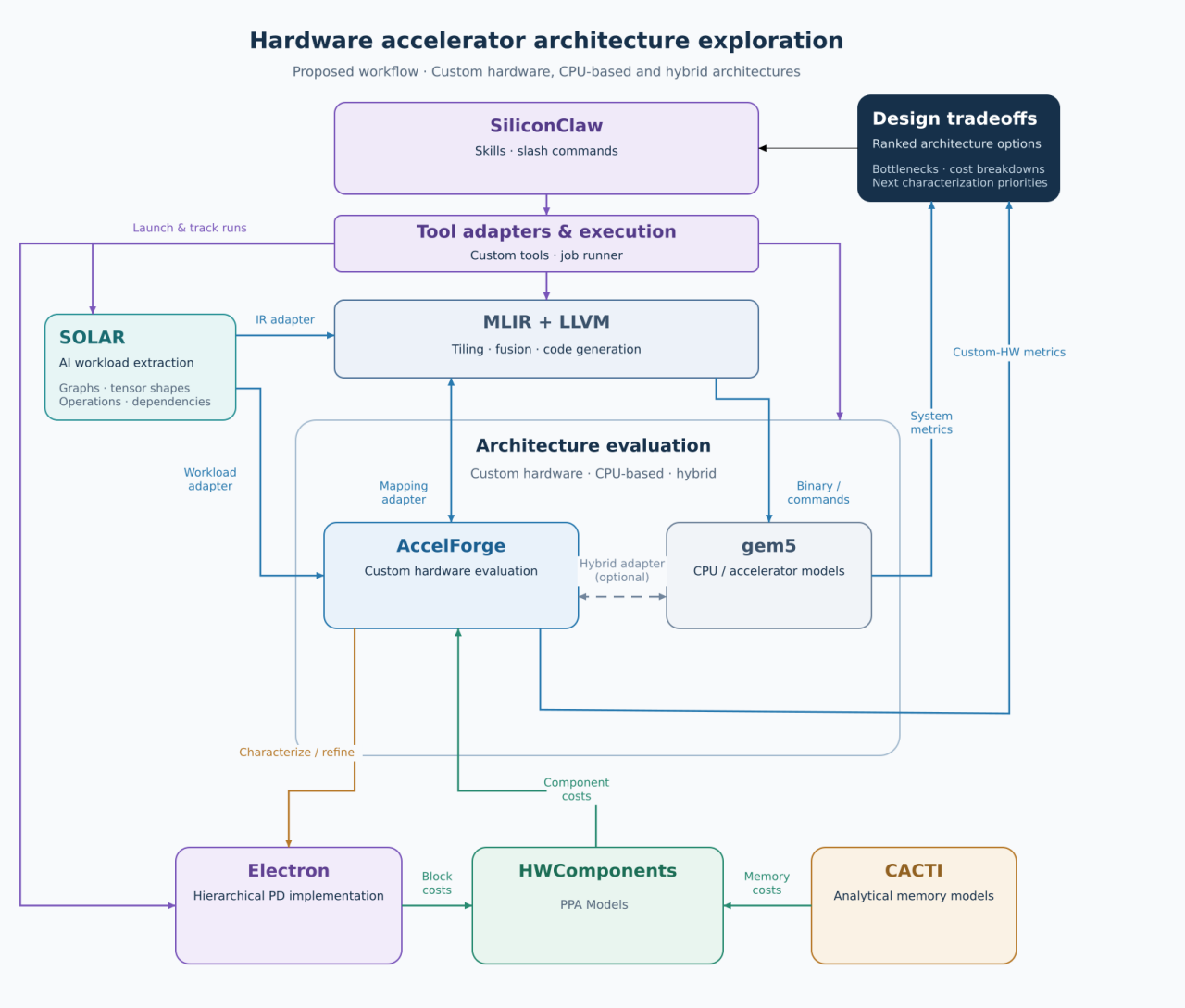
1. SOLAR: Extracts AI workload structure and computational requirements.
2. MLIR + LLVM: Provide compiler infrastructure for optimization and code generation.
3. AccelForge: Explores mappings and evaluates custom accelerator architectures.
4. gem5: Simulates CPU execution and systems with modeled accelerators.
5. HWComponents: Provides reusable component area, power, and latency models.
6. CACTI: Estimates memory area, energy, and access time.
7. Electron: Physically characterizes selected blocks and improves the component library.

SiliconClaw is an initiative I started as a fork of OpenCode to connect these tools and guide architecture exploration. Skills guide the choice of experiments and the interpretation of results, while custom commands provide entry points for common workflows. A tool-adapter and execution layer prepares inputs, launches and tracks runs, and returns the results. For example, SOLAR workload information needs to be adapted for AccelForge, while gem5 CPU simulations run programs compiled for the target CPU. Electron results can be added to HWComponents to improve future estimates.

Why do we need this?

Accelerator exploration today is spread across separate tools for workload analysis, hardware evaluation, system simulation, and physical design. The handoff between them is often manual, making it difficult to compare designs consistently or use implementation results in the next round of exploration. This framework puts those stages in a loop. SiliconClaw proposes candidates, the execution layer runs the appropriate evaluations, and the results guide the next experiment. Promising hardware blocks can receive more detailed analysis in Electron, which gradually improves the reusable component models. The same workflow can explore custom hardware, CPU-based designs, and hybrid systems, with or without a host controller.

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This project is still evolving. I am now working through the tool adapters and the compiler and simulator support needed to connect each stage reliably.

Thanks for reading,
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